

From Manual Product Entry to a Governed Commerce System

Shopify Product Architecture | Metafield Taxonomy | AI-Assisted Content | Product Operations

ROLE: PRODUCT OWNER / DIGITAL & AI SYSTEMS

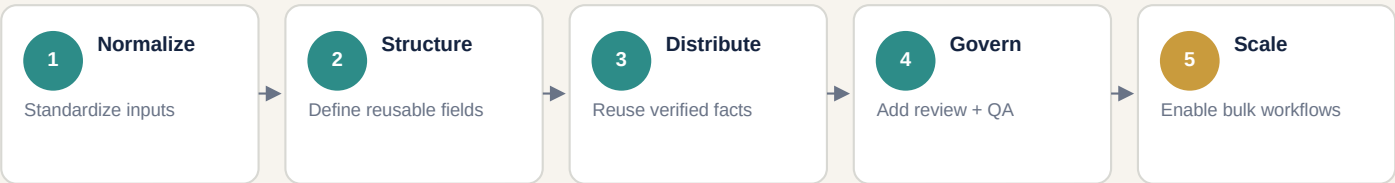
2022-2025

The operating problem

A luxury multi-brand catalog depended on inconsistent vendor data, repeated manual writing, fragmented image production, and product-by-product decisions. The catalog could continue growing; the operating model could not.



The system mandate



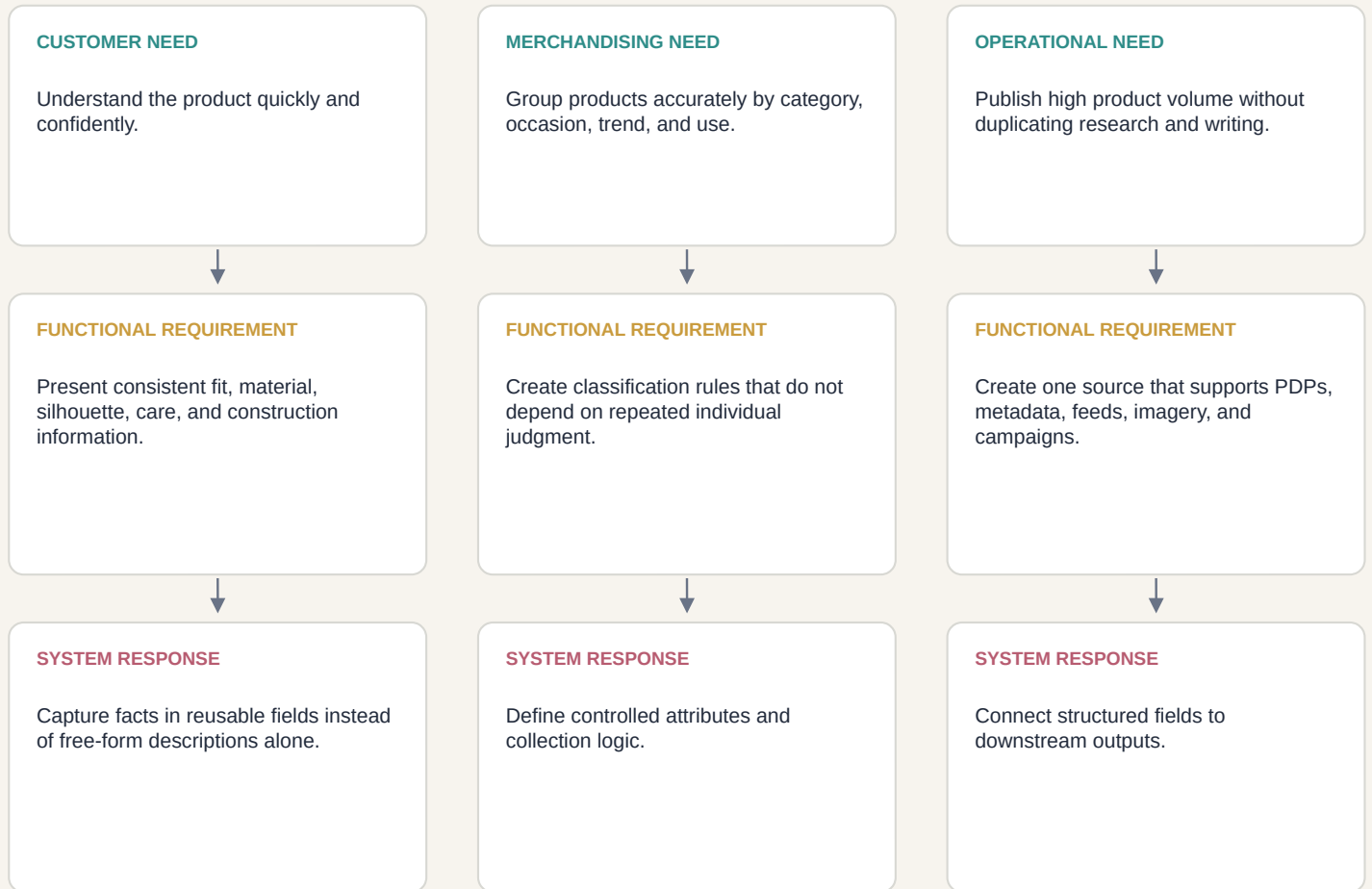
CORE DESIGN QUESTION How could verified product facts move from intake to publication without sacrificing accuracy, merchandising logic, or luxury brand standards?

CONFIDENTIALITY NOTE

All product examples, interfaces, data tables, and workflow diagrams are reconstructed and anonymized. No client admin views, customer data, logos, proprietary spreadsheets, or third-party product photography are shown.

Translating business needs into a scalable product architecture

Automation was not the starting point. The first requirement was a shared definition of what the business needed to know about every product, how that information should be verified, and where it needed to appear.



CORE REQUIREMENTS

Accuracy

Consistency

Reusability

Flexibility

Governance

Scalability

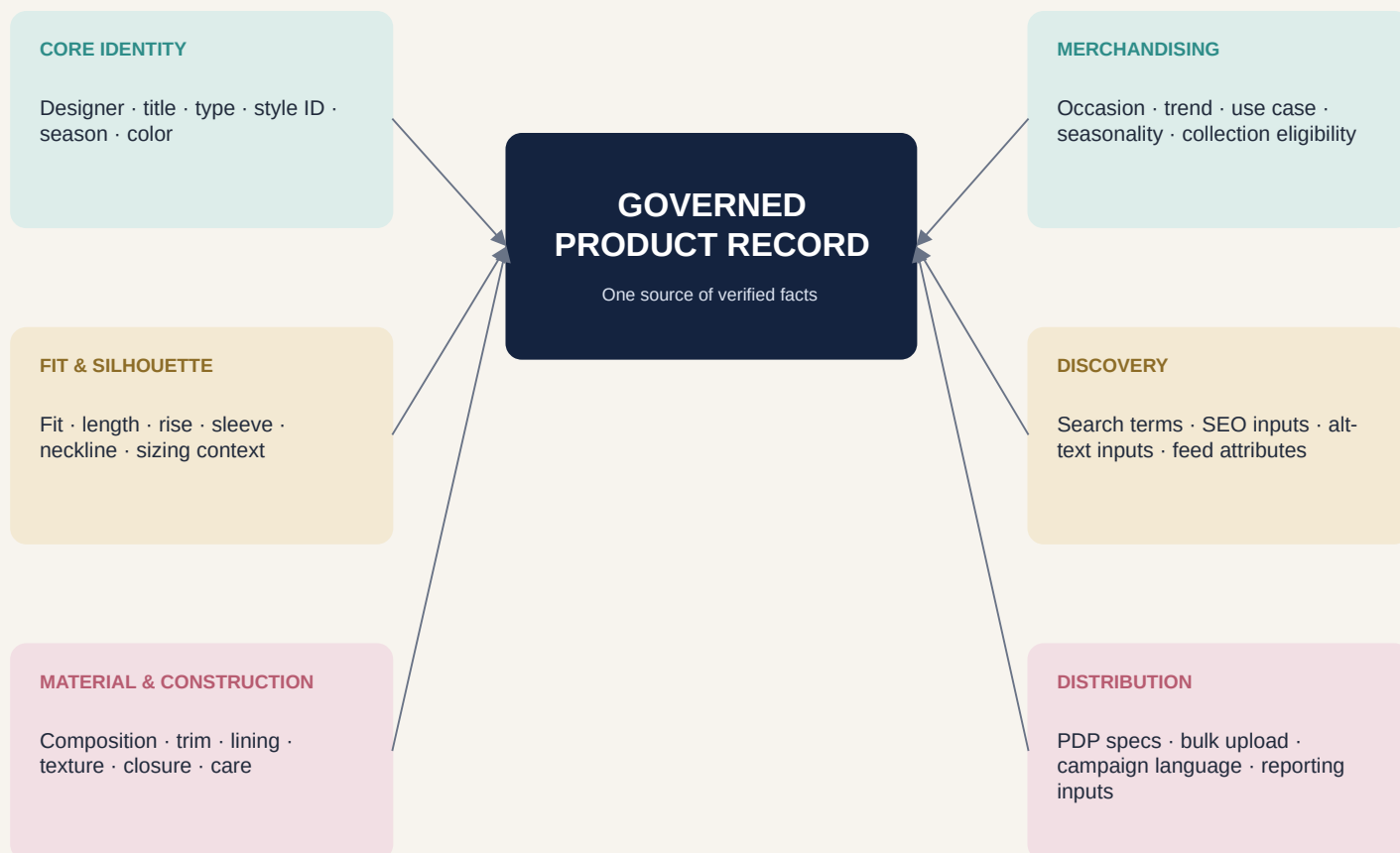
SYSTEM DESIGN PRINCIPLE

Separate verified product truth from the way that truth is presented.

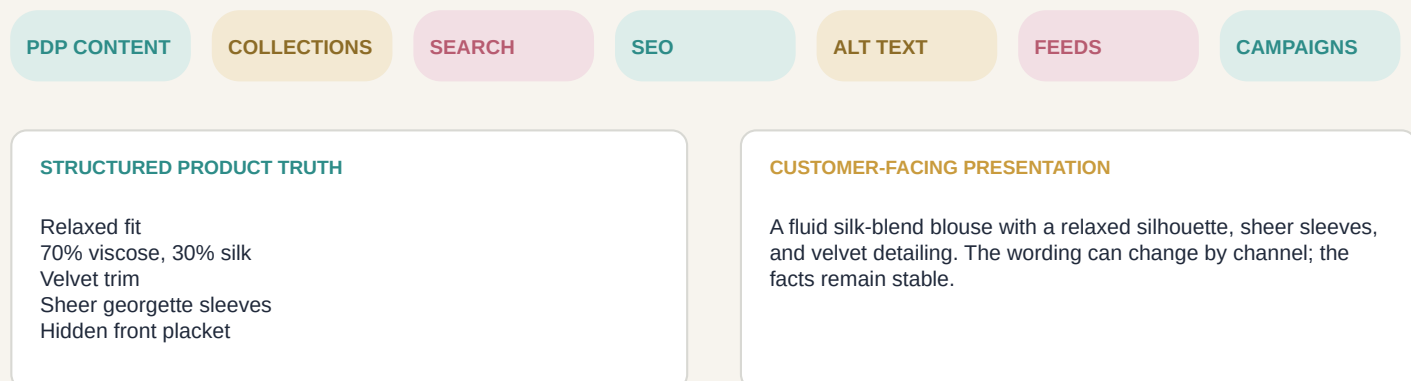
Vendor sources → Fact verification → Governed product record → PDPs · collections · search · SEO · alt text · feeds · campaigns · reporting

Building one reliable source of product truth

Critical details were separated from free-form copy and stored as reusable attributes. Customer-facing language could evolve without changing the underlying product record.



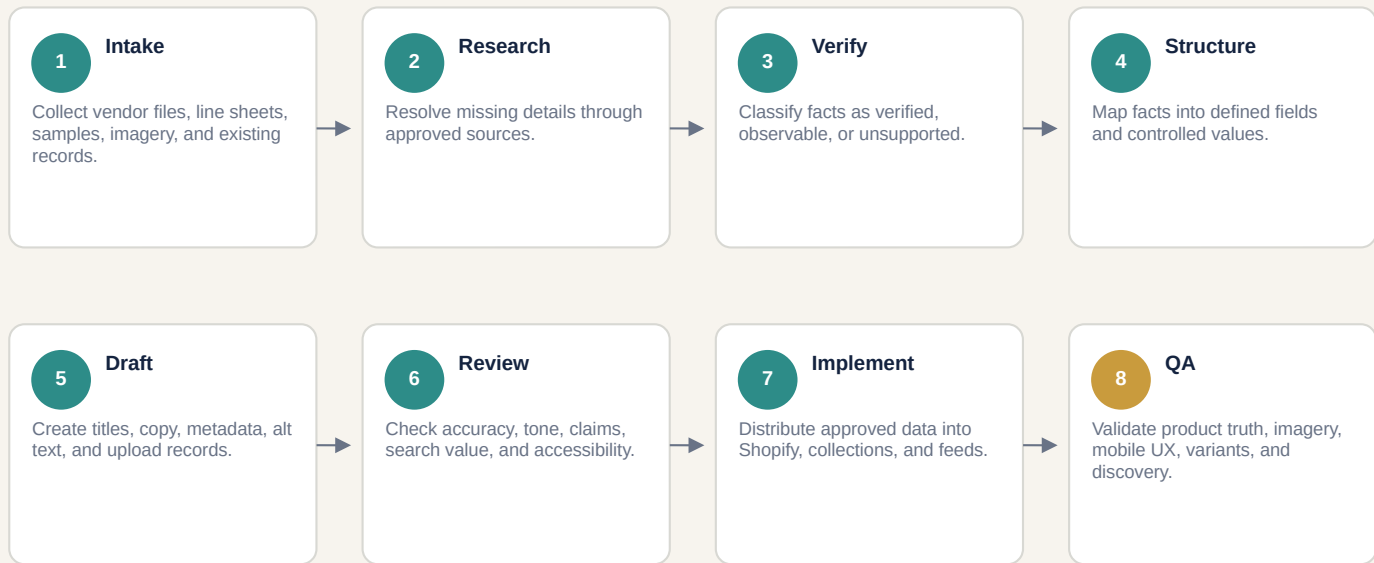
ONE FACT, MULTIPLE USES



Structure the facts that must remain consistent. Preserve flexibility where human interpretation adds customer value.

Turning fragmented inputs into a repeatable publishing system

The workflow defined where automation could accelerate production, where human judgment was required, and which checks had to be completed before a product could go live.



EVIDENCE CONFIDENCE

VERIFIED

Confirmed through vendor documentation, labels, samples, or approved sources.

OBSERVABLE

Clearly visible in imagery or the physical product but not formally documented.

UNSUPPORTED

Incomplete, contradictory, inferred, or excluded from publication.

AUTOMATION BOUNDARIES

AI + AUTOMATION ACCELERATED

Initial field mapping
Controlled-value recommendations
First-pass descriptions and metadata
Alt-text drafts
Bulk-file preparation
Repetitive transformations

HUMAN OWNERS RETAINED CONTROL

Source credibility
Fact verification
Fit and construction interpretation
Luxury editorial judgment
Image accuracy
Exceptions and final approval

Automation could accelerate a verified process. It could not replace the verification process.

Protecting product truth across every customer-facing output

Documented standards made quality transferable. Written and visual outputs were reviewed together because they had to describe the same product.

FACTS → PRESENTATION

FICTIONAL PRODUCT EXAMPLE

Governed facts

Relaxed silhouette
70% viscose, 30% silk
Velvet collar and trim
Sheer georgette sleeves
Hidden front placket
Removable neck tie

Customer-facing description

A fluid silk-blend blouse shaped with voluminous sheer sleeves and a relaxed silhouette. Velvet detailing defines the collar, while a removable tie offers another styling option.

IMAGERY STANDARDS

SILHOUETTE

Preserve real proportions, drape, volume, waist placement, and intentional asymmetry.

MATERIAL

Keep silk fluid, velvet dimensional, lace layered, and transparency believable.

CONSTRUCTION

Match seams, hems, closures, trim, panels, and front/back continuity.

PRESENTATION

Use clean backgrounds, soft frontal light, consistent crop, scale, naming, and image order.

Improve presentation without changing the product.

QUALITY-CONTROL LOOP



SYSTEM IMPROVEMENT EXAMPLES

Inaccurate cuff → construction rule

Distorted volume → silhouette rule

Over-smoothed fabric → material rule

Inconsistent terminology → controlled vocabulary

Repeated metadata edits → field-definition update

Creating infrastructure that could scale beyond individual execution

The result was not simply faster entry. It was a governed commerce system connecting product truth, operating workflows, customer experience, merchandising, and future automation.

BEFORE

FRAGMENTED PRODUCTION

Inconsistent vendor inputs
Facts buried in emails and free text
Repeated research and rewriting
Informal content and image review
Individual interpretation drove classification
Automation risked scaling inconsistency

AFTER

GOVERNED PRODUCT SYSTEM

Verified facts captured once
Controlled attributes supported reuse
Content and imagery shared standards
Review responsibilities were explicit
Corrections became system rules
Automation operated within boundaries

BUSINESS VALUE CREATED

OPERATIONAL CONSISTENCY

A repeatable path across products, designers, and source formats.

CATALOG SCALABILITY

Fewer one-off decisions as new products entered the catalog.

CUSTOMER CLARITY

More consistent, scannable, and useful product information.

MERCHANDISING CONTROL

Reusable fields for collections, classification, search, and filters.

AUTOMATION READINESS

Structured inputs, rules, and review gates for responsible AI use.

PRODUCT-LEADERSHIP TAKEAWAY

The most valuable automation was designing a reliable system that determined what could be trusted, how information should move, and where human judgment remained essential.

MY OWNERSHIP

PRODUCT + SYSTEMS

Product strategy · requirements definition · information architecture · taxonomy and metafields · workflow design · AI governance · QA standards · documentation

CAPABILITIES DEMONSTRATED

Product ownership · Shopify operations · structured content · human-in-the-loop systems · cross-functional translation · change enablement

Diagnosing Revenue Decline in a Mobile-First Commerce Ecosystem

Behavioral Analytics | Safari Mobile | Core Web Vitals | Search Visibility | Revenue Modeling

ROLE: DIGITAL DIRECTOR / PRODUCT & GROWTH STRATEGY

2024-2025

The diagnostic challenge

Revenue was declining across several digital channels at the same time, but the signals did not point to one obvious cause. Some conversion and profitability indicators were improving even as traffic and total sales contracted.

-77%

MARKETING CAPACITY

Recovery could not depend primarily on additional media spend.

MIXED

PERFORMANCE SIGNALS

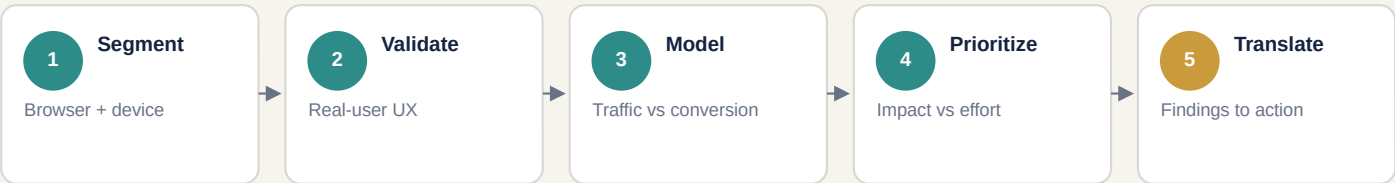
Traffic, conversion, browser, search, and margin trends moved in different directions.

SILOED

REPORTING

Search, storefront, product, campaign, and financial evidence lacked one shared diagnostic frame.

The diagnostic method



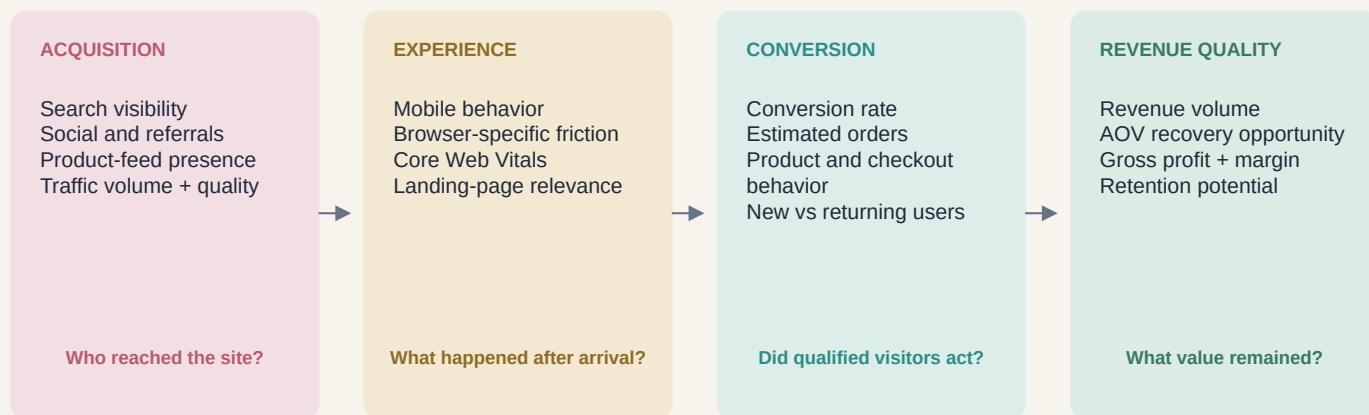
CENTRAL QUESTION Was revenue falling because the storefront was failing - or because too few qualified customers were reaching an increasingly effective experience?

CONFIDENTIALITY NOTE

All visuals are reconstructed from internal analysis and anonymized. Company identifiers, proprietary screenshots, customer-level data, campaign creative, and vendor materials are excluded. Modeled scenarios are labeled separately from actual performance.

Separating symptoms from root causes

Blended averages could support the wrong intervention. The analysis separated acquisition, experience, conversion, and revenue quality before assigning a cause or recovery action.



EVIDENCE SOURCES



DECISION LOGIC

- Where did the decline originate?** Segment by browser, device, source, and customer type.
- Was storefront friction still primary?** Use real-user performance and conversion evidence, not assumption.
- Which lever held more recovery value?** Model traffic restoration separately from conversion improvement.
- What was recoverable under constraint?** Prioritize organic visibility, feed health, retention, and product relevance.

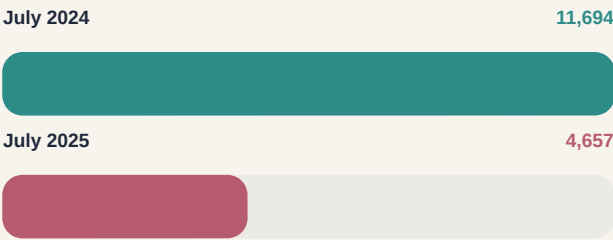
Lower revenue did not automatically mean the storefront was converting poorly.

Traffic loss overwhelmed conversion improvement

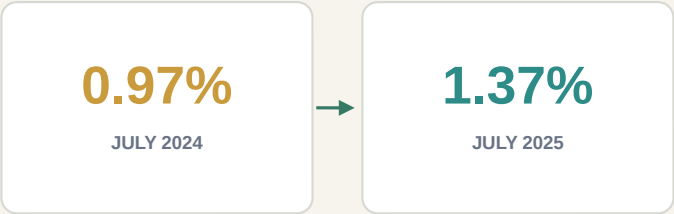
A year-over-year Safari Mobile comparison revealed that the conversion rate improved, but the available audience shrank too sharply for that gain to protect order volume or revenue.

ACTUAL PERFORMANCE

Safari Mobile sessions



Conversion rate



-60.2% TRAFFIC

+41.2% RELATIVE

ESTIMATED ORDER EFFECT

~113 ORDERS

Estimated July 2024 Safari Mobile conversions.

~64 ORDERS

Estimated July 2025 Safari Mobile conversions.

-43.8%

Estimated order decline despite the stronger conversion rate.

CONSERVATIVE REVENUE RECOVERY MODEL

ACTUAL JULY 2025

\$18.5K

Safari Mobile revenue

MODELED AT RESTORED TRAFFIC

\$46.3K

Maintaining 1.37% conversion and ~\$289 implied AOV

+\$27.9K

+151%

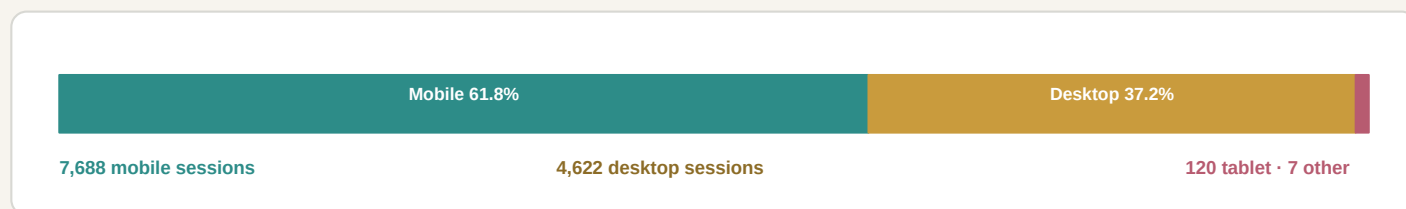
DIAGNOSTIC VERDICT

Traffic and reach loss - not conversion collapse - was the larger constrained recovery opportunity.

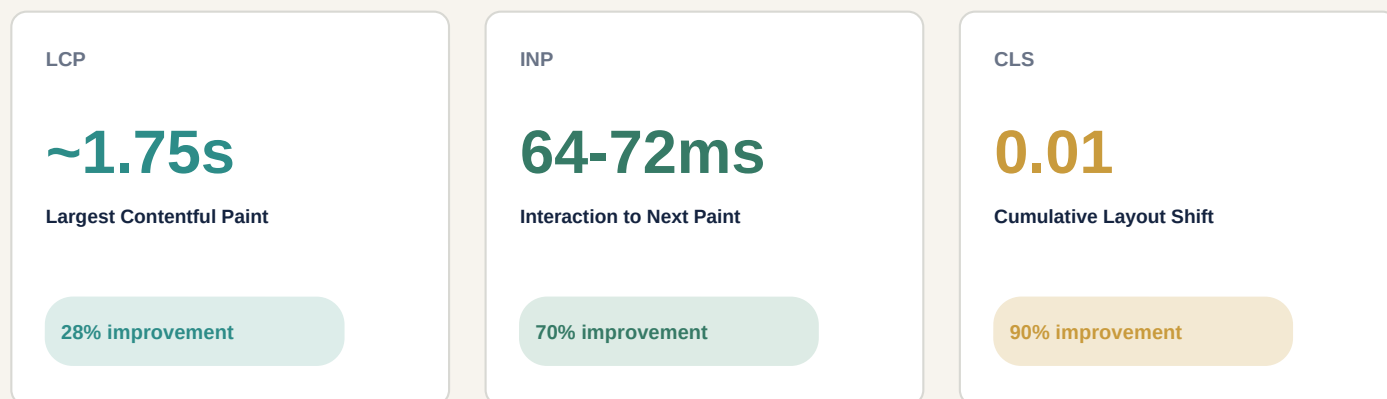
Testing whether the storefront remained the primary constraint

Real-user evidence showed a mobile-led audience and materially improved Core Web Vitals. That changed the diagnosis from “the site is still broken” to “the optimized experience is underfired by qualified traffic.”

REAL-USER DEVICE SAMPLE



CORE WEB VITALS - REAL USER P75



STATUS SHIFT

FAILING → **FULLY GREEN**

Performance work remained important, but the evidence no longer supported treating speed as the only or primary explanation for revenue decline.

WHAT THE EVIDENCE CHANGED

PRIOR ASSUMPTION Lower revenue likely reflected a storefront that was still too slow or unstable.	VALIDATED FINDING The optimized mobile experience performed within strong real-user thresholds.	RECOVERY IMPLICATION Shift priority toward qualified traffic, product visibility, and landing relevance.
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Turning fragmented findings into low-cost, high-leverage action

With marketing spend reduced by 77%, the strategy emphasized recoverable structural issues: product visibility, feed health, organic discovery, owned-channel quality, and evidence-led mobile merchandising.

PRIORITIZATION MATRIX



RECOVERY WORKSTREAMS

- 1 Search + feed health**
Diagnose product-feed and Shopping visibility issues; strengthen product eligibility and discovery.
- 2 Product architecture**
Use structured attributes, collections, PDP language, and metadata to improve relevance.
- 3 Owned audience quality**
Separate total profiles from active subscribers; use suppression, engagement, and retention signals.
- 4 Mobile-first merchandising**
Prioritize Safari-heavy behavior, landing quality, hierarchy, and product relevance.
- 5 Measurement cadence**
Track channel, browser, conversion, margin, and recovery assumptions separately.

IMPLEMENTED VS. MODELED

Implemented: performance optimization, product and content architecture, browser/device analysis, search and feed diagnosis, lifecycle analysis.
Modeled or recommended: restored-traffic revenue scenario, broader search recovery, expanded low-cost acquisition and retention roadmap.

PRIORITIZATION PRINCIPLE

Repair the structural paths that qualified customers use before buying more traffic to the same broken paths.

Replacing metric overload with an evidence-led recovery system

The work changed the business question from “Why are sales down?” to a set of answerable decisions about acquisition, experience, conversion, customer value, and recoverable infrastructure.

THREE DIAGNOSTIC CONCLUSIONS

1 / TRAFFIC LOSS DOMINATED

Safari Mobile traffic fell 60.2%. The 41.2% relative conversion gain could not offset the smaller audience.

2 / EXPERIENCE IMPROVED

Real-user Core Web Vitals moved from failing to fully green, reducing the evidence for speed as the sole cause.

3 / RECOVERY WAS STRUCTURAL

Search visibility, product relevance, feed health, retention, and qualified traffic offered higher-leverage paths under constraint.

DECISION-MAKING VALUE

ROOT-CAUSE CLARITY

Separated channel loss from conversion and technical performance.

REVENUE MODELING

Quantified a conservative + \$27.9K monthly Safari Mobile opportunity.

RESOURCE FOCUS

Redirected limited capacity toward recoverable structural levers.

MEASUREMENT DISCIPLINE

Kept actuals, estimates, and recommended work visibly distinct.

CROSS-FUNCTIONAL TRANSLATION

Connected analytics to product, UX, SEO, lifecycle, and operations.

STRATEGIC TAKEAWAY

The strongest recovery opportunity was not a single redesign or campaign. It was a coordinated system for restoring qualified visibility while preserving an experience that had already become more effective.

MY OWNERSHIP

ANALYSIS + STRATEGY

Diagnostic framing · segmentation · Core Web Vitals interpretation · revenue scenario modeling · search/feed investigation · retention and audience analysis · prioritization

CAPABILITIES DEMONSTRATED

Behavioral analytics · mobile-first strategy · product and growth leadership · decision modeling · cross-channel systems thinking · executive communication